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Stochastic semi-discretization for linear stochastic delay differential equations

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Summary
An efficient numerical method is presented to analyze the moment stability and stationary behavior of linear stochastic delay differential equations. The method is based on a special kind of discretization technique with respect to the past effects. The resulting approximate system is a high dimensional linear discrete stochastic mapping. The convergence properties of the method is demonstrated with the help of the stochastic Hayes equation and the stochastic delayed oscillator.

KEYWORDS
moment stability, stationary solution, stochastic dynamical systems, time delay

1 | INTRODUCTION

There are several models in engineering and biology, which lead to delay differential equations (DDE), eg, the models of machine tool vibrations,[1-3] delayed control loops,[4] traffic dynamics,[5] predator-prey systems,[6] or neural networks.[7] During the investigation of such delayed dynamical systems, stochastic effects are usually neglected, although stochastic excitations often influence the behavior of these systems. The noise may appear not only as an external excitation but also in the coefficients of the state variables.

In case of the construction of deterministic models that approximate stochastic systems, the mean values of the measured system parameters are used to describe the process, whereas the measured variation is considered as the noise in the measurement.[8] This mean value–based approach is valid for linear systems only, and even in these cases, its result may be misleading regarding the stationary behavior. A noise excitation can lead not only to change in stability properties but it can also cause a so-called autonomous stochastic resonance or coherence resonance.[9,10]

One of the reasons why the stochastic effects in delayed systems are usually neglected is the complexity of the mathematical tools necessary to investigate their dynamic properties. Another reason is that the accessible methods are often not applicable effectively. One of these methods is based on the analysis of trajectories calculated with time-domain Monte Carlo (MC) simulation in order to determine stability. One needs a large amount of trajectories to properly characterize the system; moreover, the applied numerical methods may also influence the results of the stability analysis.[11] When conducting simulations, one has to be careful since the stochastic terms in the equations are not integrated with respect to $\mathrm{d}t$, but rather to a process which has increments proportional to $\sqrt{\mathrm{d}t}$.

Another approach is to use small perturbation of a critical parameter to investigate the stationary properties of the selected solution.[12,13] An alternative method is to approximate the stationary probability density function (sPDF) of the phase angle in the polar-coordinate representation of the stochastic DDE (SDDE) using Fokker-Planck equation.[14] This approach leads to a nonlinear deterministic system of equations for the dominant characteristic exponent, the average phase angle velocity, and the sPDF of the phase angle, which can be solved in an iterative way or by using the multidimensional bisection method.[14,15] The two latter methods have the weaknesses of not only being intuitive and problem specific but usually applicable only for small degree-of-freedom dynamical systems.

There are several definitions for the stability properties of linear stochastic systems. One definition is the *almost sure exponential stability*,[16-18] which investigates the mean decay rate of the exponential envelope of a selected solution of the stochastic system. If a solution of a system is almost sure exponentially stable, then even if there are finite number of bursts in the trajectories, the solution converges to an equilibrium state on average, while it may not have a stationary solution because of the bursts. However, even if the solution is a stationary solution, it can produce such stationary motions that prevent the effective use of applications developed by deterministic models due to the stochastic coherence resonance. Such a system appears in the model of regenerative machine tool vibrations, where the cutting tool vibrates due to the stochastic force excitation originated in the nonhomogeneous workpiece material.[10] If this system has the aforementioned bursts, the mechanical model predicts unacceptable machining conditions.

A conservative stability definition is the so-called *moment stability*.[19] The analysis of both the first and second moment evolution is sufficient[19] to determine if a stochastic system has a stationary solution. If the stochastic system has a stationary solution,[19] it can be used to decide whether the size of its envelope is below a prescribed bound for the given application. The evolution of the first moment defines the mean dynamics, the second moment provides information also about the average deviations around the mean dynamics, whereas the higher order moment dynamics do not contain additional information regarding stability.

Since the moment stability is the most conservative stability definition, it is the best candidate for engineering applications. This paper aims to introduce a simple-to-use general and efficient method to analyze moment stability based on semi-discretization. The terminology semi-discretization is used here in the same sense as in case of partial differential equations,[20] that is, only the delayed section of the state variable is discretized, whereas the time derivatives are not. This is a kind of intermediate discretization as opposed to the full discretization, where time derivatives are also approximated, eg, by finite difference schemes. In the paper of Elbeyli et al,[21] the basics of the moment mapping and the stability investigation method are laid down, but as opposed to the title of that paper, a full discretization scheme is applied based on the Euler-Maruyama method. Due to its slow convergence properties and its high computational demand, it is not optimal for delayed problems especially with multiple state variables. In the present study, the numerical algorithm is derived based on semi-discretization,[20] which has higher order convergence properties than the full-discretization–based method mentioned above.

In Section 2, the class of the stochastic equations in question is introduced. In Section 3, the linear stochastic mapping[21] is defined in a general form. In Section 4, a constructive example is shown to generate the mapping matrices. Finally, in Section 5, the convergence properties of the proposed higher order mapping are presented. The result of the method is compared to the second moment of the solution of a simple, analytically solvable, scalar first-order stochastic Hayes equation with delay. The convergence properties are further demonstrated by using the complex problem of the stochastic delayed oscillator.

To encourage the use of the proposed method, an open source Julia[22] package is provided under the name of *StochasticSemiDiscretizationMethod.jl*; this tool supports the investigation of the moment stability and stationary solution of linear stochastic delayed systems.

2 | LINEARIZATION OF STOCHASTIC DELAY DIFFERENTIAL EQUATIONS

Consider the following autonomous deterministic DDE:

$$\dot{\mathbf{x}}_t = \mathbf{a}\left(\mathbf{x}_t, \mathbf{x}_{t-\tau}\right) \tag{1}$$

with initial condition $\mathbf{x}_t = \boldsymbol{\phi}(t)$, $t \in [-\tau, 0]$, and $\mathbf{a} \in C^1\left(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d\right)$ right-hand side. The dot refers to differentiation w.r.t. the time t, while $t \mapsto \mathbf{x}_t \in \mathbb{R}^d$ is the d-dimensional deterministic state variable. Note that $\mathbf{x}_t$ refers to the state $\mathbf{x}$ at the time instant t as opposed to the notation used generally in DDEs defined with the shift transformation w.r.t. the delay.[23,24] Assume that there exists a trivial solution $\mathbf{x}^*$ (equilibrium state) of the system. Without the loss of generality, one can set $\mathbf{x}^* \equiv \mathbf{0}$ such that $\mathbf{a}\left(\mathbf{0}, \mathbf{0}\right) = \mathbf{0}$. Consider that the conditions are fulfilled, which guarantee that the differential equation linearized around the trivial solution $\mathbf{x}^*$ exhibits the same stability properties in the neighborhood of the equilibrium state as the original nonlinear equation.[19,25] If there exist the coefficient matrices $\mathbf{A}$, $\mathbf{B} \in \mathbb{R}^{d\times d}$, such that

$$|\mathbf{a}(\boldsymbol{\xi},\boldsymbol{\eta}) - \mathbf{A}\boldsymbol{\xi} - \mathbf{B}\boldsymbol{\eta}| = o\left(|\boldsymbol{\xi}| + |\boldsymbol{\eta}|\right), \tag{2}$$

then the investigation of the stability properties of

$$\dot{\mathbf{x}}_t = \mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{x}_{t-\tau} \tag{3}$$

can be used to prove the local stability of the trivial solution $\mathbf{x}^*$ of the nonlinear DDE (1).

To analyze the effect of stochastic perturbation on the stability properties, Equation (1) is extended as follows:

$$\dot{\mathbf{x}}_t = \mathbf{a}\left(\mathbf{x}_t, \mathbf{x}_{t-\tau}\right) + \mathbf{b}\left(\mathbf{x}_t, \mathbf{x}_{t-\tau}\right)\Gamma_t + \boldsymbol{\sigma}\Gamma_t, \tag{4}$$

where function $\mathbf{b} \in C^1\left(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d\right)$ has also the property of $\mathbf{b}(\mathbf{0},\mathbf{0}) = \mathbf{0}$, $\boldsymbol{\sigma} \in \mathbb{R}^d$ is a constant vector representing the intensity of the additive noise, and $t \mapsto \Gamma_t \in \mathbb{R}$ is a one-dimensional Langevin[26] force excitation. This time, $\mathbf{x}_t$ denotes the d-dimensional stochastic process. The goal is to investigate whether the process $\mathbf{x}_t$ stays in a bounded region of the trivial solution $\mathbf{x}^*$ of Equation (1). For this purpose, the definition of the first and second moment stability is used[19]:

$$\lim_{t\to\infty} |\langle \mathbf{x}_t \rangle| = 0, \qquad \lim_{t\to\infty} \left|\left\langle \mathbf{x}_t\mathbf{x}_t^\top \right\rangle\right| \le \varepsilon, \qquad 0 < \varepsilon < \infty \qquad \forall \mathbf{x}_t = \boldsymbol{\phi}(t),\ t \in [-\tau, 0]. \tag{5}$$

Similarly to a deterministic case, it is assumed that, for matrices $\boldsymbol{\alpha}, \boldsymbol{\beta} \in \mathbb{R}^{d\times d}$, the condition

$$|\mathbf{b}(\boldsymbol{\xi},\boldsymbol{\eta}) - \boldsymbol{\alpha}\boldsymbol{\xi} - \boldsymbol{\beta}\boldsymbol{\eta}| = o\left(|\boldsymbol{\xi}| + |\boldsymbol{\eta}|\right) \qquad \boldsymbol{\xi},\boldsymbol{\eta} \in \mathbb{R}^d \text{ and } |\boldsymbol{\xi} + \boldsymbol{\eta}| \to 0 \tag{6}$$

holds. Then, the linear SDDE of the form

$$\dot{\mathbf{x}}_t = \left(\mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{x}_{t-\tau}\right) + \left(\boldsymbol{\alpha}\mathbf{x}_t + \boldsymbol{\beta}\mathbf{x}_{t-\tau} + \boldsymbol{\sigma}\right)\Gamma_t \tag{7}$$

can be used to prove moment stability and to approximate the stationary behavior of the process defined by the nonlinear SDDE equation (4). An appropriate way to construct the matrices $\mathbf{A}$, $\mathbf{B}$, $\boldsymbol{\alpha}$, and $\boldsymbol{\beta}$ is to use the first-order Taylor expansion of $\mathbf{a}(\boldsymbol{\xi},\boldsymbol{\eta})$ and $\mathbf{b}(\boldsymbol{\xi},\boldsymbol{\eta})$ at the fix point $\mathbf{x}^* = \mathbf{0}$.

Since the integration of the Langevin force leads to the Wiener process,[26] the appropriate approach is to rewrite Equation (7) into incremental form

$$\mathrm{d}\mathbf{x}_t = \left(\mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{x}_{t-\tau}\right)\mathrm{d}t + \left(\boldsymbol{\alpha}\mathbf{x}_t + \boldsymbol{\beta}\mathbf{x}_{t-\tau} + \boldsymbol{\sigma}\right)\mathrm{d}W_t, \tag{8}$$

where $t \mapsto W_t \in \mathbb{R}$ is the Wiener process. The Wiener increments have the following properties for the expected value and variance: $\mathbb{E}\left(\mathrm{d}W_t\right) = 0$ and $\mathbb{E}\left(\mathrm{d}W_t\mathrm{d}W_s\right) = \mathrm{d}t \cdot \mathbb{1}(t = s)$, where the indicator function $\mathbb{1}(.)$ gives 1 or 0 if the argument is true or false, respectively. When solving stochastic differential equations (SDE), one can use two definitions for the stochastic integral, namely, the Stratonovich integral and the Itô integral. Since the transformation from one to the other is well defined,[26] only the Itô definition is considered here during the derivations.

3 | STOCHASTIC MAPPING

In order to numerically investigate the stationary solution of the SDDE (8), the state space of the continuous functions is reduced to a discrete one (see the sketch in Figure 1), and the SDDE (8) can be approximated as a discrete stochastic mapping of the form

$$\mathbf{y}_{n+1} = (\mathbf{F} + \mathbf{G})\,\mathbf{y}_n + \mathbf{g}. \tag{9}$$

The discretized state space vector $\mathbf{y}_n \in \mathbb{R}^{(p+1)d}$ at the time $n\Delta t$ is defined as introduced by Insperger and Stépán[20,27]:

$$\mathbf{y}_n = \left[\mathbf{x}_{n\Delta t}^\top, \mathbf{x}_{(n-1)\Delta t}^\top, \ \ldots, \mathbf{x}_{(n-p)\Delta t}^\top\right]^\top, \tag{10}$$

where $n \in \mathbb{N}$ identifies the discrete time instant, Δt is the resolution of the time variable, and p is a properly chosen integer to include the delayed term such that the time delay $\tau \le p\Delta t$. Note that in case of a single discrete delay τ with zeroth-order approximation, $\tau = p\Delta t$ is a convenient choice.

In Equation (9), the deterministic mapping matrix $\mathbf{F}$ is calculated from $\mathbf{A}$ and $\mathbf{B}$, the stochastic perturbation mapping matrix $\mathbf{G}$ is calculated using $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, and the stochastic additive noise vector $\mathbf{g}$ is calculated from $\boldsymbol{\sigma}$. Furthermore, it is assumed that $\mathbf{G}$ is independent of $\mathbf{F}$ and $\mathbf{y}_n$, but not from $\mathbf{g}$, also $\mathbf{G}$ and $\mathbf{g}$ are (discrete-)time dependent stochastic variables.

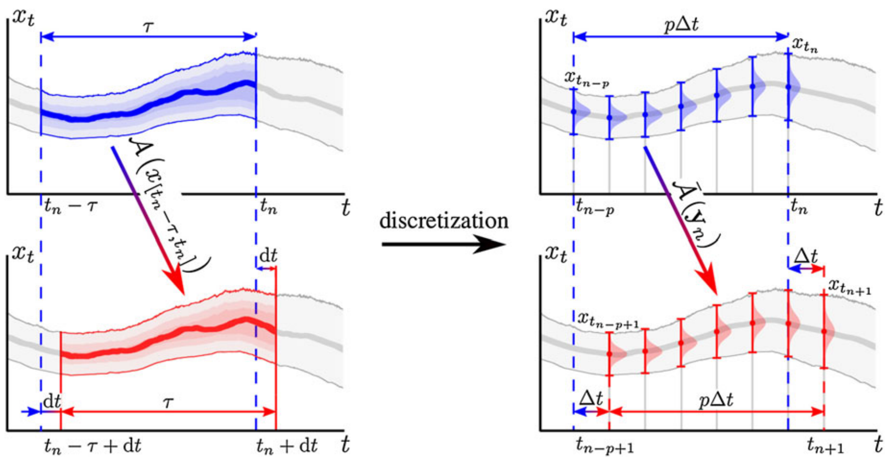

FIGURE 1 Discretization of the stochastic process, where $\mathcal{A} : x_{[t_n-\tau,t_n]} \mapsto x_{[t_n+\mathrm{d}t-\tau,t_n+\mathrm{d}t]}$ represents the operator which maps the function x forward for an infinitesimal time dt. On the right side, $\bar{A} : \mathbf{y}_n \mapsto \mathbf{y}_{n+1}$ represents the discretized approximation of the operator $\mathcal{A}$, which maps the discretized function x_t stored in the vector $\mathbf{y}$ (see Equation (10)) at the finite time Δt forward

Since the stochastic properties of $\mathbf{G}$ and $\mathbf{g}$ are not changing in time, as an abuse of notation, the time dependence of these variables is omitted ($\mathbf{G} := \mathbf{G}_n, \mathbf{g} := \mathbf{g}_n$).

To approximate the first moment evolution of the SDDE (8), the expected value of Equation (9) can be used. Since $\mathbf{F}$ is defined to be deterministic and $\mathbf{G}$ describes the stochastic variations around $\mathbf{F}$, the first moment mapping is simply

$$\langle \mathbf{y}_{n+1} \rangle = \mathbf{F} \langle \mathbf{y}_n \rangle, \tag{11}$$

where the notation $\langle . \rangle$ is used to represent the ensemble average.

Equation (11) gives the underlying discrete deterministic delayed system of Equation (8). The stability of this system can be characterized by the spectral radius

$$\rho(\mathbf{F}) = \max_{z} \{\mathrm{abs}(z),\ z \in \mathbb{C} : \det\ (\mathbf{F} - z\mathbf{I}) = 0\} \tag{12}$$

of the mapping matrix $\mathbf{F}$.[20,27] Note that if the stochastic effects are neglected during modeling, that is, the stochastic components are omitted from the governing Equation (7) or (8), one ends up with the same deterministic mapping as Equation (11); the stochastic effects do not influence the mean dynamics of a linear system.

When dealing with the moment stability of a linear SDDE, the second moment dynamics has to be investigated. For this purpose, the definition of the squared process is needed:

$$\mathbf{y}_{n+1}\mathbf{y}_{n+1}^{\top} = (\mathbf{F}+\mathbf{G})\,\mathbf{y}_n\mathbf{y}_n^{\top}(\mathbf{F}+\mathbf{G})^{\top} + (\mathbf{F}+\mathbf{G})\,\mathbf{y}_n\,\mathbf{g}^{\top} + \mathbf{g}\,\mathbf{y}_n^{\top}(\mathbf{F}+\mathbf{G})^{\top} + \mathbf{g}\mathbf{g}^{\top}, \tag{13}$$

which can be rewritten using Einstein's notation as

$$y_{i,n+1}y_{j,n+1} = \left(F_{il}F_{jk} + G_{il}G_{jk} + F_{il}G_{jk} + G_{il}F_{jk}\right) y_{l,n}y_{k,n} + \left(G_{ik}g_j + G_{jk}g_i + F_{ik}g_j + F_{jk}g_i\right) y_{n,k} + g_i g_j. \tag{14}$$

The expected value of the squared process describes the evolution of the second moment:

$$\langle y_{n+1,i}y_{n+1,j}\rangle = \left(F_{il}F_{jk} + \langle G_{il}G_{jk}\rangle\right)\langle y_{n,l}y_{n,k}\rangle + \langle G_{ik}g_j + G_{jk}g_i\rangle\langle y_{n,k}\rangle + \langle g_i g_j\rangle. \tag{15}$$

During the derivation of Equation (15) for the second moment mapping, the following properties are used[19,26]:

$$\langle F_{il}\rangle = F_{il}, \quad \langle G_{il}\rangle = 0, \quad \langle F_{il}G_{jk}\rangle = F_{il}\langle G_{jk}\rangle = 0. \tag{16}$$

Furthermore, the value of G_{il} is independent of $y_{n,j}$. Introducing the notation

$$\mathbf{M}(n) = \left\langle \mathbf{y}_n\mathbf{y}_n^{\top}\right\rangle \in \mathbb{R}^{(p+1)d\times(p+1)d} \tag{17}$$

for the second moment matrix, one can rewrite Equation (15) in the following form:

$$M_{ij}(n+1) = \left(F_{il}F_{jk} + \langle G_{il}G_{jk}\rangle\right) M_{lk}(n) + \langle G_{ik}g_j + G_{jk}g_i\rangle\langle y_{n,k}\rangle + \langle g_i g_j\rangle. \tag{18}$$

To distinguish stochastic and deterministic quantities, the convention introduced by Arnold[19] is used: the subscript n refers to stochastic dependence, whereas n in parentheses refers to deterministic dependence. Instead of the matrix-to-matrix mapping representations of the second moment mapping in Equations (13)-(18), it is convenient to represent it as a vector-to-vector mapping. One possible arrangement is to take each diagonals of the second moment matrix:

$$\mathbf{m}(n) := \left[M_{11}(n), M_{22}(n), \ldots, M_{12}(n), M_{23}(n), \ldots, M_{1,(p+1)d}(n)\right]^{\top}, \tag{19}$$

where the symmetric elements of matrix $\mathbf{M}(n)$ are only considered once. Using the second moment vector introduced in Equation (19), the mapping in Equation (18) has the form

$$\mathbf{m}(n+1) = \mathbf{H}\mathbf{m}(n) + \mathbf{C}\langle \mathbf{y}_n \rangle + \mathbf{c}, \tag{20}$$

where the coefficient matrices $\mathbf{H}$, $\mathbf{C}$, and $\mathbf{c}$ are generated using the rules defined in (18). An example for their construction is presented in Appendix C where the matrices $\mathbf{F}$ and $\mathbf{G}$ have the sizes $(p+1)d = 3$ and $(p+1)d = 4$.

The stability of the second moment process is determined by the spectral radius of $\mathbf{H}$: if $\rho(\mathbf{H}) > 1$, then the second moment of the process defined in Equation (7) is unstable, and if $\rho(\mathbf{H}) < 1$, then it is stable and the process has a bounded stationary solution.

Note that the first moment process $\langle \mathbf{y}_n \rangle$ appears in Equation (20) as an additive excitation, independent of $\mathbf{m}(n)$. According to Arnold,[19] it is shown that the second moment might be stable ($\rho(\mathbf{H}) < 1$) only if the first moment is stable ($\rho(\mathbf{F}) < 1$). Therefore, if the squared process is stable, then the stationary excitation from the first moment process disappears: $\lim_{n\to\infty}\langle \mathbf{y}_n \rangle = 0$, consequently the determination of matrix $\mathbf{C}$ is needed neither for the stability investigation nor for the calculation of the stationary second moment. The additive term $\mathbf{c}$ is a bounded quantity originating from the stochastic integral of the additive process denoted by $\boldsymbol{\sigma}\mathrm{d}W_t$ in Equation (8), and the stationary squared process $\mathbf{m}_{\mathrm{st}}$ can be calculated as

$$\mathbf{m}_{\mathrm{st}} = (\mathbf{I} - \mathbf{H})^{-1}\mathbf{c}. \tag{21}$$

The above described method makes it possible to investigate the effect of the parameters in SDDE (8). Accordingly, the so-called stability charts can be constructed in the space of these parameters based on the spectral radius of the coefficient matrix $\mathbf{H}$ of the second moment vector $\mathbf{m}(n)$. In the meantime, the second moment $\mathbf{m}_{\mathrm{st}}$ of the stationary solution can also be approximated in the stable domain.

To implement these steps, only the deterministic mapping matrix $\mathbf{F}$, the stochastic perturbation mapping matrix $\mathbf{G}$, and the stochastic additive noise vector $\mathbf{g}$ must be calculated. The subsequent section introduces a possible choice of the approximation method to be used during these calculations.

4 | GENERATION OF MAPPING MATRICES

There are several methods to generate the mapping matrices detailed in Section 3. One possibility is to approximate the mapping matrix as the matrix form of the Euler-Maruyama method–based full discretization.[21] This means the full discretization of the process in Equation (8), which has first-order convergence in the deterministic part and only halfth-order convergence in the stochastic part w.r.t. the number of the discretized points.[11] This leads to high computational cost for an accurate second moment analysis.

As an alternative and more efficient way of computations, the zeroth-order semi-discretization method is presented in details in the next subsection, whereas its higher order version is given in Appendix B.2.

4.1 | Zeroth-order semi-discretization

In order approximate the mapping matrices $\mathbf{F}$ and $\mathbf{G}$ and vector $\mathbf{g}$ in Equation (9), the delayed state in Equation (8) is approximated to be constant for the sufficiently small time period Δt

$$\mathrm{d}\mathbf{x}_s = \left(\mathbf{A}\mathbf{x}_s + \mathbf{B}\mathbf{x}_{t_n-\tau}\right)\mathrm{d}s + \left(\boldsymbol{\alpha}e^{\mathbf{A}(s-t_n)}\mathbf{x}_{t_n} + \boldsymbol{\beta}\mathbf{x}_{t_n-\tau} + \boldsymbol{\sigma}\right)\mathrm{d}W_s. \text{ where } s \in [t_n, t_{n+1}), \quad n = 0, 1, 2\ldots, \tag{22}$$

which results constant external excitation for $s \in [t_n, t_{n+1})$. The multiplicative stochastic term has to be approximated since there exists no explicit analytic solution for the general linear SDE since the matrices $\mathbf{A}$ and $\boldsymbol{\alpha}$ are generally not exchangeable (see details in the work of Arnold[19]). There are two simple ways to approximate the multiplicative term at present time: as a constant $\boldsymbol{\alpha}\mathbf{x}_{t_n}$ or as if it behaves according to the expected value $\boldsymbol{\alpha}e^{\mathbf{A}s}\mathbf{x}_{t_n}$ of the nondelayed equation. In Equation (22), the latter approach is chosen for reasons detailed in Appendix B.1.

With the above assumptions, one can write SDDE (8) into the approximating difference equation

$$\mathbf{x}_{t+\Delta t} = (\mathbf{P}+\mathbf{Q})\,\mathbf{x}_t + (\mathbf{R}+\mathbf{S})\,\mathbf{x}_{t-\tau} + \mathbf{w}, \tag{23}$$

where the deterministic coefficient matrices can be calculated explicitly from the deterministic part (the coefficients of ds) of Equation (22) according to[27]

$$\mathbf{P} = e^{\mathbf{A}\Delta t}, \quad \mathbf{R} = \mathbf{A}^{-1}(\mathbf{P}-\mathbf{I})\,\mathbf{B}. \tag{24}$$

Since Equation (22) is a linear SDE with additive noise, the stochastic coefficient matrices **Q** and **S**, and the stochastic additive vector **w** originated in the coefficients $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, $\boldsymbol{\sigma}$ in Equation (22) are represented[19] by the following Itô integrals:

$$\mathbf{Q} = \int_{t_n}^{t_{n+1}} e^{\mathbf{A}(t_{n+1}-s)}\boldsymbol{\alpha}e^{\mathbf{A}(s-t_n)}\mathrm{d}W_s, \qquad \mathbf{S} = \int_{t_n}^{t_{n+1}} e^{\mathbf{A}(t_{n+1}-s)}\boldsymbol{\beta}\mathrm{d}W_s, \tag{25}$$

$$\mathbf{w} = \int_{t_n}^{t_{n+1}} e^{\mathbf{A}(t_{n+1}-s)}\boldsymbol{\sigma}\mathrm{d}W_s. \tag{26}$$

Although these matrices cannot be calculated in closed form based on these formulas, their first and second moments can be evaluated numerically as given in the subsequent section. Here, the time dependence of the stochastic matrices **Q** and **S** and vector **w** are omitted since their stochastic property does not change in time. Choosing the discretization resolution in Equation (10) for the specific case $p = \tau/\Delta t$, the mapping defined in Equation (9) can be generated with the mapping matrices and additive noise vector

$$\mathbf{F} = \begin{pmatrix} \mathbf{P} & \mathbf{0} & \dots & \mathbf{0} & \mathbf{R} \\ \mathbf{I} & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{I} & \mathbf{0} \end{pmatrix}, \quad \mathbf{G} = \begin{pmatrix} \mathbf{Q} & \mathbf{0} & \dots & \mathbf{0} & \mathbf{S} \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \end{pmatrix}, \quad \mathbf{g} = \begin{pmatrix} \mathbf{w} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{pmatrix}, \tag{27}$$

respectively. Note that matrix **F** is the same already defined in the work of Insperger and Stépán,[27] whereas **G** can be built in a similar way, but its elements are stochastic integrals. This choice of mapping satisfies the assumption given for the mapping defined in (9) for the derivation of the moment mappings. The stochastic perturbation mapping matrix **G** defined in Equation (27) using (25) is independent of $\mathbf{y}_n$ since, for any function $f(.)$, the conditional expected value is equal to the expected value: $\langle f(\mathbf{G})|\mathcal{F}_{n\Delta t}\rangle = \langle f(\mathbf{G})\rangle$. It means that the stochastic matrix process **G** does not depend on the history of the process up to $t = n\Delta t$ denoted with the filtration $\mathcal{F}_{n\Delta t}$. This is due to the fact that the stochastic integrals are evaluated on the interval $(n\Delta t, (n+1)\Delta t]$.

The structure of the second moment mapping matrix **H** in Equation (20) is presented in Appendix C for $p = 3$. For its calculation, one only needs the expected value of the product of the above integrals in Equations (18)-(20), which can be approximated, eg, with the help of MC simulations. The definition of the mapping matrices **F** and **G** is not limited to the zeroth-order semi-discretization but can also be built using higher order semi-discretization as detailed in B.2. The term "higher order" refers to the order of the Lagrange polynomial used to approximate the delayed terms: during zeroth-order semi-discretization a constant is used, when applying first-order semi-discretization a linear approximation is fit, during the second-order approximation a parabola is applied, and so on.

In the subsequent section, an efficient approach is presented to approximate the expected value of the products of the stochastic coefficients **G** and **g** to be used in the construction of mapping (20).

4.2 | Evaluation of the expected values

To calculate the second moment mapping matrix **H**, there are terms originating in the stochastic matrix **G**. The nonzero elements of matrix **G** can be written as a stochastic integral of the deterministic functions $s \mapsto f_{G,ij}(s)$ $i,j = 1, 2, \dots, (p+1)d$, defined by

$$\mathbf{Q} = \int_{t_n}^{t_{n+1}} e^{\mathbf{A}(t_{n+1}-s)}\boldsymbol{\alpha}e^{\mathbf{A}s}\mathrm{d}W_{s+n\Delta t} =: \int_{t_n}^{t_{n+1}} \begin{bmatrix} f_{G,11}(s) & \dots & f_{G,1d}(s) \\ \vdots & \ddots & \vdots \\ f_{G,d1}(s) & \dots & f_{G,dd}(s) \end{bmatrix} \mathrm{d}W_s, \tag{28}$$

$$\mathbf{S} = \int_{t_n}^{t_{n+1}} e^{\mathbf{A}(t_{n+1}-s)}\boldsymbol{\beta}\mathrm{d}W_s =: \int_{t_n}^{t_{n+1}} \begin{bmatrix} f_{G,1\ pd+1}(s) & \dots & f_{G,1\ pd+d}(s) \\ \vdots & \ddots & \vdots \\ f_{G,d\ pd+1}(s) & \dots & f_{G,d\ pd+d}(s) \end{bmatrix} \mathrm{d}W_s, \tag{29}$$

when $f_{G,il}(s) \equiv 0$ where $G_{il} = 0$ in Equation (27). The elements of the vector $\mathbf{g}$ can be written similarly by means of

$$\mathbf{w} = \int_{t_n}^{t_{n+1}} e^{\mathbf{A}(t-s)}\boldsymbol{\sigma}\mathrm{d}W_s =: \int_{t_n}^{t_{n+1}} \begin{bmatrix} f_{g,1} \\ \vdots \\ f_{g,p} \end{bmatrix} \mathrm{d}W_s, \tag{30}$$

when $f_{g,i}(s) \equiv 0$ where $g_i = 0$ in Equation (27):

The Itô isometry[26] is used to evaluate the expected value of the products of the elements of matrix $\mathbf{G}$ and the vector $\mathbf{g}$, as defined in Equation (18).

$$\langle G_{il}G_{jk}\rangle = \left\langle \int_{t_n}^{t_{n+1}} f_{G,il}(s)\mathrm{d}W_s \int_{t_n}^{t_{n+1}} f_{G,jk}(s)\mathrm{d}W_s \right\rangle = \int_{t_n}^{t_{n+1}} f_{G,il}(s) f_{G,jk}(s)\,\mathrm{d}s, \tag{31}$$

$$\langle G_{ik}g_j\rangle = \left\langle \int_{t_n}^{t_{n+1}} f_{G,ik}(s)\mathrm{d}W_s \int_{t_n}^{t_{n+1}} f_{g,j}(s)\mathrm{d}W_{s+} \right\rangle = \int_{t_n}^{t_{n+1}} f_{G,ik}(s) f_{g,j}(s)\,\mathrm{d}s, \tag{32}$$

$$\langle g_i g_j\rangle = \left\langle \int_{t_n}^{t_{n+1}} f_{g,i}(s)\mathrm{d}W_s \int_{t_n}^{t_{n+1}} f_{g,j}(s)\mathrm{d}W_s \right\rangle = \int_{t_n}^{t_{n+1}} f_{g,i}(s) f_{g,j}(s)\,\mathrm{d}s \tag{33}$$

During the numerical construction of the mapping matrix $\mathbf{H}$ in Equation (20), the elements of $\mathbf{G}$ are not calculated directly; only the arguments of the integrals in Equation (31)-(33) are computed at discrete points in the time intervals evaluated by an integration scheme (eg, trapezoidal or a quadrature). These are stored in a vectorized form, and then during the second moment matrix calculation, they are multiplied with each other (according the definition in Equation (18)), then weighted and summed according the used integration scheme. This calculation has been implemented in our Julia package (see the Conclusion in Section 6).

5 | TESTING THE METHOD

In the previous sections, the construction of the proposed method has been given to investigate the stationary behavior of the linear SDDE defined in Equation (8). The method is based on the analysis of the approximate discrete system Equation (20) based on Equations (24)-(33). In the following subsections, the convergence properties of the method are investigated numerically. The first test example is a scalar first-order delayed equation for which an analytic solution exists, so one can check whether the results gained by the discrete mapping converge to this exact solution. In case of the second example, the advantages of the semi-discretization method are even more characteristic in terms of the convergence of both the spectral radius and the stationary solution compared to the full discretization.

5.1 | Test case 1: stochastic Hayes equation

The Hayes equation is one of the simplest scalar examples to investigate the behavior of DDEs.[23,28] We use its stochastic version

$$\mathrm{d}x_t = Ax_t\mathrm{d}t + \beta x_{t-1}\mathrm{d}W_t + \sigma\mathrm{d}W_t, \tag{34}$$

with initial condition

$$x_t = \phi_t, \quad t \in [-1, 0], \tag{35}$$

where $A,\ \beta,\ \sigma \in \mathbb{R}$. Note that the delay can be chosen to be 1 without loss of generality.

5.1.1 | Analytic solution

The stability properties of Equation (34) can be determined analytically. The solution of Equation (34) can be written in the following form[29]:

$$x_t = e^{At}x_0 + \int_0^t e^{A(t-s)}\beta x_{t-1} + e^{A(t-s)}\sigma\mathrm{d}W_s. \tag{36}$$

The mean dynamics is

$$\langle x_t \rangle = e^{At} x_0, \tag{37}$$

which converges to 0 if the proportional parameter $A < 0$. This means that the system is first moment stable:

$$\lim_{t\to\infty} \langle x_t \rangle = \lim_{t\to\infty} \langle x_{t-1} \rangle = 0. \tag{38}$$

Still, it is necessary to further investigate the stationary behavior of the process in Equation (34); the second moment $M(t) = \langle x_t x_t \rangle$ of the solution (36) needs to be evaluated using the Itô isometry[26]

$$M(t) = e^{2At} M_0 + \frac{\sigma^2}{2A}\left(e^{2At} - 1\right) + \int_0^t \beta^2 e^{2A(t-s)} M(t-1) + 2e^{2A(t-s)} \sigma\beta \langle x_{t-1} \rangle \mathrm{d}s. \tag{39}$$

A large enough time instant t_0 is chosen to have the solution converged to the stationary second moment $M_{\mathrm{st}} = \lim_{t\to\infty} M(t)$. If $0 \ll t_0 \ll t$, then

$$M_{\mathrm{st}} = -\frac{\sigma^2}{2A} + \int_{t_0}^t \beta^2 e^{2A(t-s)} \mathrm{d}s M_{\mathrm{st}}, \tag{40}$$

from where

$$M_{\mathrm{st}} = -\frac{\sigma^2}{2A + \beta^2}. \tag{41}$$

Since M_{st} has to be nonnegative and σ^2 is always positive, the second moment is stable only if

$$A < -\frac{\beta^2}{2}, \tag{42}$$

which is a stricter condition than $A < 0$ for the first moment stability in Equation (38).

Note that these analytical results and also the decay rate of the transient part can be calculated by means of Itô's lemma, too. The squared process of the stochastic Hayes Equation (34) assumes the form[26]

$$\mathrm{d}x_t^2 = \left(2Ax_t^2 + \beta^2 x_{t-1}^2 + \sigma^2 + 2\beta\sigma x_{t-1}\right) \mathrm{d}t + (\beta x_{t-1} + \sigma)\,\mathrm{d}W_t. \tag{43}$$

Taking the expected value of the squared process Equation (43) and transforming the resulting increment equation into differential form lead to the following deterministic DDE:

$$\dot{M}(t) = 2AM(t) + \beta^2 M(t-1) + 2\beta\sigma\langle x_{t-1}\rangle + \sigma^2. \tag{44}$$

Note that, after the substitution of Equation (38) into Equation (44), the calculated stationary solution is equivalent to the result described in Equation (41). Furthermore, it is possible to determine the characteristic exponents of the perturbed second moment $\tilde{M}(t) = M(t) - M_{\mathrm{st}}$ and the following homogeneous DDE[23] is obtained:

$$\dot{\tilde{M}}(t) = 2A\tilde{M}_t + \beta^2 \tilde{M}(t-1). \tag{45}$$

Using the exponential trial solution

$$\tilde{M}(t) = ce^{\lambda t}, \tag{46}$$

one arrives at the following characteristic equation:

$$\lambda - 2A - \beta^2 e^{-\lambda} = 0. \tag{47}$$

The characteristic exponent λ can be partitioned as $\lambda = \gamma + \mathrm{i}\omega$, $\gamma, \omega \in \mathbb{R}$, where γ and ω are related to the decay rate and to the angular frequency of the transient vibration, respectively. To determine the stability boundaries, one has to substitute $\gamma = 0$ into Equation (47) and separate the real and complex terms and find the roots of the corresponding implicit equations[23]

$$2A + \beta^2 \cos\omega = 0, \tag{48}$$

$$\omega + \beta^2 \sin\omega = 0. \tag{49}$$

The analytic solution of Equations (48)-(49) is the following parametric curves in case of Hopf bifurcation ($\omega \neq 0$):

$$A = \frac{\omega}{2\tan(\omega)}, \qquad \beta = \pm\sqrt{-\frac{\omega}{\sin(\omega)}}, \qquad \omega \in \left[(2k-1)\pi, 2k\pi\right), k \in \mathbb{Z}, \tag{50}$$

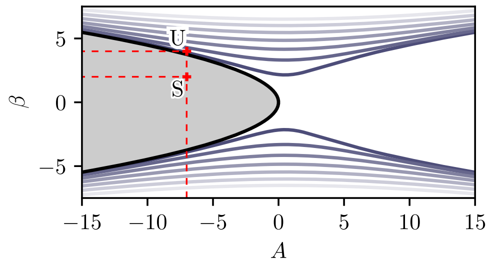

FIGURE 2 Stable area (gray) and the bifurcation curves defined by Equations (48)-(51) [Colour figure can be viewed at wileyonlinelibrary.com]

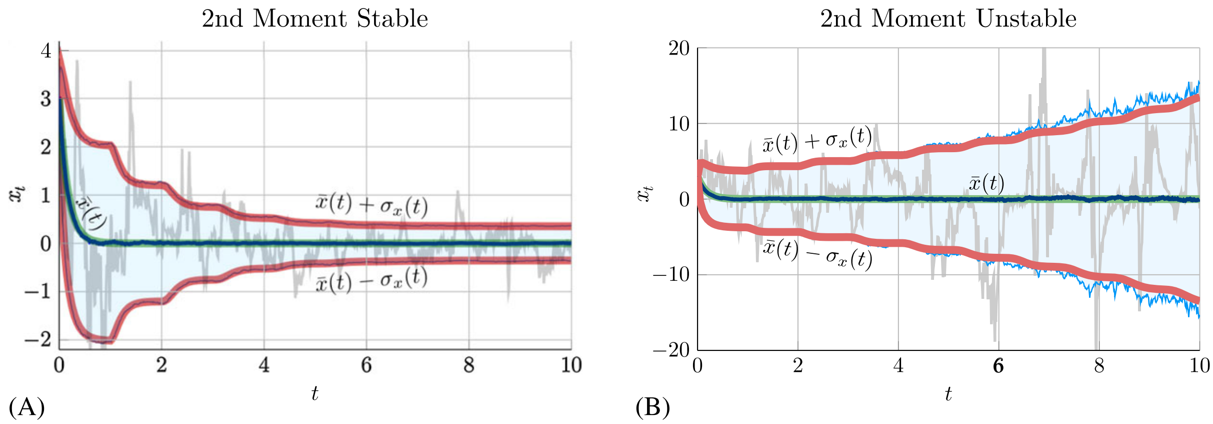

FIGURE 3 The first moment (green curve) and its envelopes (red curves) calculated by means of Equations (37) and (44), respectively, are compared to the statistical evaluation of MC simulations: the dark blue line represents the mean and the light blue lines and areas represents the standard deviation of 10^4 realizations. Panel (A) shows a stable case ($A = -6, \beta = 2$, denoted by parameter point "S" in the stability chart of Figure 2). Panel (B) shows a second moment unstable case ($A = -6, \beta = 4$, denoted with point "U" in Figure 2). $\bar{x}(t) = \langle x_t \rangle$, $\sigma_x(t) = \sqrt{M(t) - \bar{x}(t)^2}$, and initial function $\phi_t \equiv 3$, $t \in [-1, 0]$. The gray trajectories represent optionally selected stochastic realizations

while in case of fold bifurcation ($\omega = 0$), it gives

$$A = -\frac{\beta^2}{2}. \tag{51}$$

Figure 2 presents the critical parameter lines defined by Equation (50)-(51). Note that the stability boundary is defined by a fold-type bifurcation only (denoted by the shaded region). Assuming that the dominant characteristic root is real ($\omega = 0$), it is possible to write γ as the function of A and β

$$\gamma = 2A + W_0\left(\beta^2 e^{-2A}\right), \tag{52}$$

which characterizes the decay rate[30] of the second moment process $M(t)$. In this formula, W_0 is the main branch of the Lambert W function,[31] which is by definition the function $W(z)$ which satisfies $W(z)e^{W(z)} = z$.

5.1.2 | MC simulations of the stochastic Hayes equation

The above described results are compared to numerical simulations. In Figure 3, the first and second moment evolutions are obtained by the deterministic Equations (37) and (44), and these are compared to the statistical evaluation of MC simulations, where the expected value and the standard deviation of the trajectories are presented on the basis of 10^4 realizations of full discretization of the SDDE (34). It can be seen that the deterministic solution (using the default DDE solver in the Julia package *DifferentialEquations.jl*[32] for Equation (44)) matches well to the mean values and deviations of the MC simulations. Note that the analytic method gives almost everywhere smooth curves for the moment evolutions, while the MC simulations give only somewhat noisy curves and with orders of magnitude higher computation time. The noisy curve for the deviation evolution can be observed especially in Figure 3B, where the stochastic process is second moment unstable.

5.1.3 | Convergence of the stochastic semi-discretization

When the semi-discretization method is applied, the mapping described in Equation (18) is constructed numerically. For the M_{11} element of this mapping for order $q = 0$, the details of this calculation are given in Appendix D and its equivalence to the corresponding discretization of the analytical solution in Equation (44) is also shown.

With the help of Equations (52), (41), and (51), the convergence of the stochastic semi-discretization method is investigated by means of the spectral radius of the second moment mapping matrix $\mathbf{H}$, the stationary second moment, and the stability chart. Since there is no deterministic delayed term in the stochastic Hayes equation (34), the convergence of the spectral radius of the first moment is not analyzed here (for details, see the work of Insperger and Stépán[27]); in this case, the semi-discretization method provides the analytic solution, while the full discretization converges only in first order.

A properly chosen discretization has to satisfy the following criterion:

$$\lim_{p\to\infty} \rho(\mathbf{H})^p = e^{\gamma}. \tag{53}$$

The expression e^{γ} describes the change of the amplitude in one delay period, while the spectral radius $\rho(\mathbf{H})$ only characterizes it within the numerical time step $\Delta t = 1/p$. To compare them, one has to use the pth power of the spectral radius. To characterize the convergence of the method, the relative error

$$\delta_{\rho,2}(p) = \frac{\left|e^{\gamma} - \rho(\mathbf{H})^p\right|}{e^{\gamma}} \tag{54}$$

between the numerical result and the analytic solution is used.

Figure 4A shows that the convergence rate of the full-discretization method[21] and the proposed semi-discretization methods of different order is the same w.r.t. p, and the errors $\delta_{\rho,2}(p)$ differ only in a constant multiplier. In this scalar case, the semi-discretization method provides an order of magnitude more accurate approximation than the full-discretization method. This is important since the number of elements of the second moment mapping matrix $\mathbf{H}$ is proportional to p^4, and the zeroth-order semi-discretization method achieves 1% accuracy at $p = 50$, whereas the full-discretization method cannot achieve this even with $p = 250$.

To investigate the convergence of the stationary second moment, the following relative error is defined:

$$\delta_{\mathrm{st}}(p) = \frac{\left|M_{\mathrm{st}} - M_{\mathrm{st},11}\right|}{M_{\mathrm{st}}}. \tag{55}$$

Note that, in this scalar case, all the diagonal elements of the second moment matrix $\mathbf{M}(n)$ in the stationary solution $\mathbf{M}_{\mathrm{st}}$ are the same. Therefore, in Equation (55), only the first element $M_{\mathrm{st},11} = \langle x_t x_t \rangle$ of the stationary second moment vector $\mathbf{m}_{\mathrm{st}}$ from Equation (21) is compared to the analytic solution M_{st} (Equation (41)).

In Figure 4B, the better convergence of the stochastic semi-discretization method can be observed again. The different order methods show a slightly smaller error although the rate of convergence is almost the same. Furthermore, the relative error for the spectral radius and the stationary second moment show similar tendency.

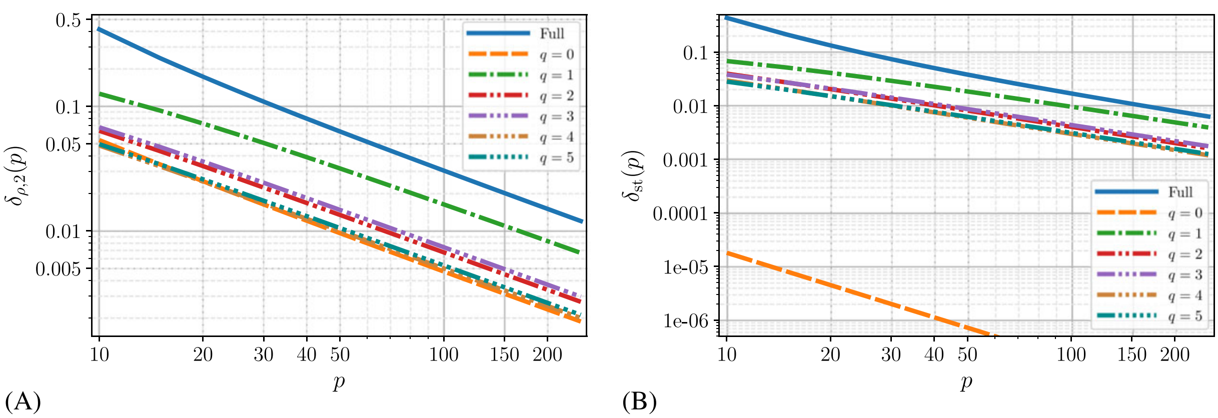

FIGURE 4 Convergence of the spectral radius of (A) the second moment mapping matrix and (B) the stationary second moment, for the stochastic Hayes Equation (34) for different order q of semi-discretization and the full-discretization method,[21] using the parameters $A = -6$, $\beta = 2$, $\sigma = 1$, $e^{\gamma} \approx 0.3640$, and $M_{\mathrm{st}} = 0.125$. The order q refers to the order of the Lagrange polynomial used to approximate the delayed terms

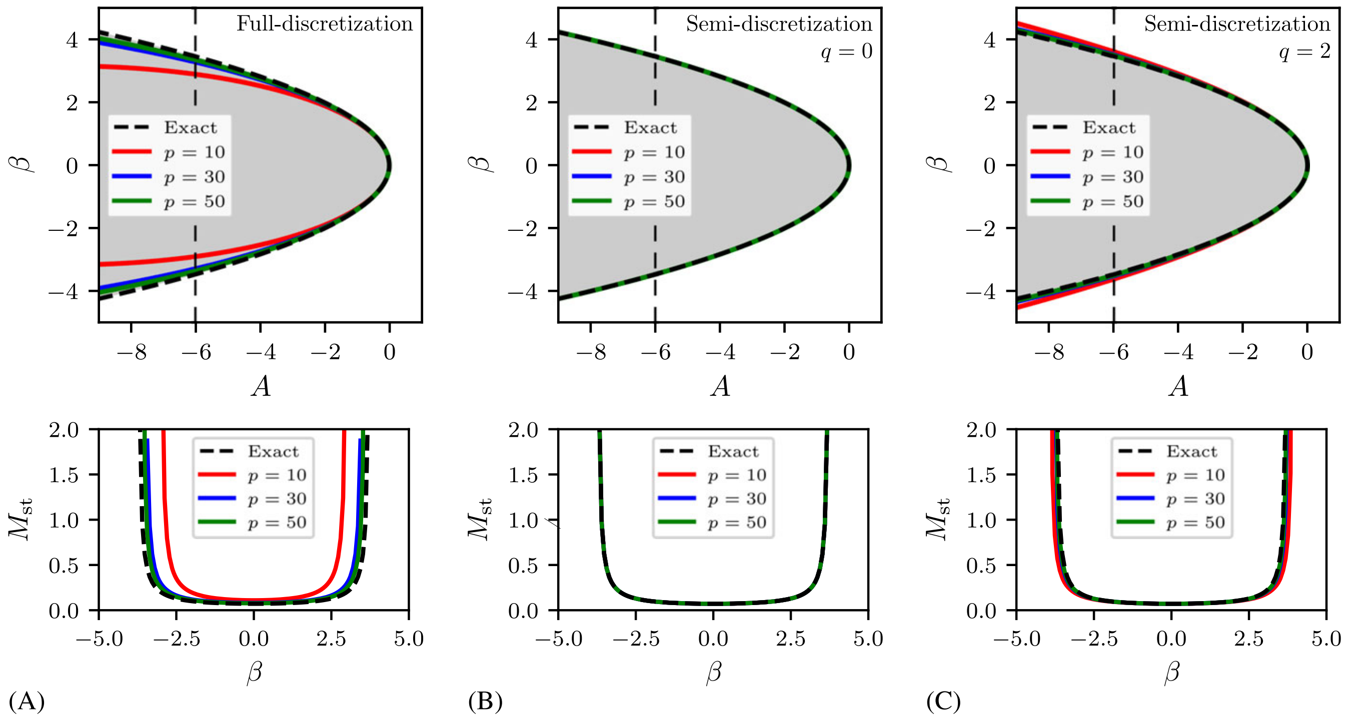

FIGURE 5 Convergence of the stability chart (upper panel) and the stationary second moment (lower panel) using (A) full discretization, (B) zeroth-order semi-discretization, and (C) second-order semi-discretization. The order q refers to the order of the Lagrange polynomial used to approximate the delayed terms. The vertical dashed lines in the stability charts represent the parameter $A = -6$ along which the stationary second moments were calculated

In Figure 5, the numerically calculated stability boundaries, defined by the spectral radius of the discretized system Equation (12), are compared to the stability boundary determined analytically from Equation (51). The stability boundary $\rho(H) = 1$ of the discretized system is determined with the help of the multidimensional bisection method proposed by Bachrathy and Stépán.[15] It can be seen that the smaller the value A in the discrete system is, the worse the approximation of the analytic stability chart is, so in the general case, the use of higher order method is useful to reach a certain desired accuracy.

At the moment, we could not find a mathematical explanation for the extremely good convergence of the zeroth-order semi-discretization compared to the higher order ones. This might be due to the special simple mathematical form of the Hayes equation. This is the reason why the method is tested on a more general problem in the next subsection.

5.2 | Test case 2: stochastic delayed oscillator

The stochastic delayed oscillator of the form

$$\dot{x}_t = v_t \tag{56}$$

$$\dot{v}_t + 2\zeta v_t + A x_t = B x_{t-2\pi} + (\alpha x_t + \beta x_{t-2\pi} + \sigma)\,\Gamma_t \tag{57}$$

is an essential mathematical model in many engineering problems like machine tool vibrations.[9,10] This can be transformed into Equation (8) with

$$\mathbf{x}_t = \begin{bmatrix} x_t \\ v_t \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 0 & 1 \\ -A & -2\zeta \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ B & 0 \end{bmatrix}, \quad \boldsymbol{\alpha} = \begin{bmatrix} 0 & 0 \\ \alpha & 0 \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} 0 & 0 \\ \beta & 0 \end{bmatrix}, \quad \boldsymbol{\sigma} = \begin{bmatrix} 0 \\ \sigma \end{bmatrix}, \quad \tau = 2\pi. \tag{58}$$

In this multidimensional case, the explicit solution like Equation (36) for the Hayes equation is not available.[19] In Figure 6, two examples are shown for the evolution of the first and second moments together with an individual realization of the process in Equations (56)-(58). The cases of stable and unstable second moments show good agreement between the statistical analysis of independent MC simulations and the spectral radius and the stationary solution computed with the stochastic semi-discretization.

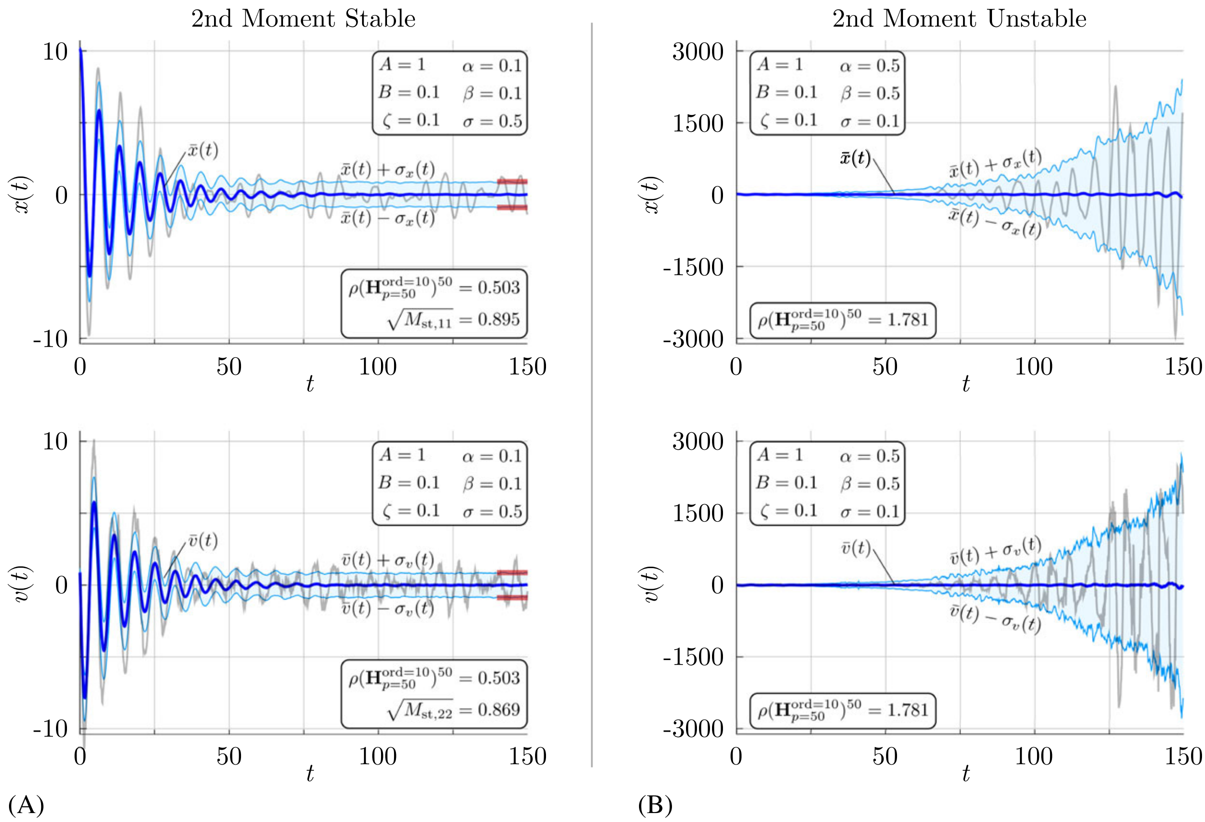

FIGURE 6 First moment stable processes are shown for stable second moment in panel (A) and for unstable second moment in panel (B). The gray lines represent typical realizations of the process. The dark blue lines show the mean, whereas the light blue lines and areas show the standard deviation of 10^4 realizations obtained by MC simulations. The red lines in panel (A) represent the stable stationary second moment calculated with stochastic semi-discretization

For comparison, the reference values of the stationary second moment $\mathbf{m}_{\mathrm{st}}$ and the spectral radii of the first moment mapping matrix $\mathbf{F}$ and second moment mapping matrix $\mathbf{H}$ are calculated using the 10*th*-order semi-discretization method with as high resolution p as our computational capacity allowed:

$$\rho_1^{\mathrm{ref}} = \rho\left(\mathbf{F}_{p=200}\right)^{200}, \tag{59}$$

$$\rho_2^{\mathrm{ref}} = \rho\left(\mathbf{H}_{p=200}\right)^{200}, \tag{60}$$

$$\mathbf{m}_{\mathrm{st}}^{\mathrm{ref}} = \left(\mathbf{I} - \mathbf{H}_{p=130}\right)^{-1}\mathbf{c}_{p=130}. \tag{61}$$

Selecting these reference values, the relative errors δ are defined as a function of the resolution parameter p:

$$\delta_{\rho,1}(p) = \frac{\left|\rho_1^{\mathrm{ref}} - \rho(\mathbf{F})^p\right|}{\left(\rho_1^{\mathrm{ref}}\right)}, \tag{62}$$

$$\delta_{\rho,2}(p) = \frac{\left|\rho_2^{\mathrm{ref}} - \rho(\mathbf{H})^p\right|}{\left(\rho_2^{\mathrm{ref}}\right)}, \tag{63}$$

$$\delta_{\mathrm{st}}(p) = \frac{\left|M_{\mathrm{st},11}^{\mathrm{ref}} - M_{\mathrm{st},11}\right|}{M_{\mathrm{st},11}^{\mathrm{ref}}}. \tag{64}$$

In Figures 7, 8, and 9, the convergence properties of the spectral radii of the first and the second moment mapping matrices and, similarly, those of the stationary second moment can be observed, respectively.

Introduce the convergence rate r_c by fitting of

$$\delta(p) = Kp^{-r_c} \tag{65}$$

for $p > 50$ onto the error functions showed in Figures 7, 8, and 9. The corresponding numerical values of r_c are given in Table 1.

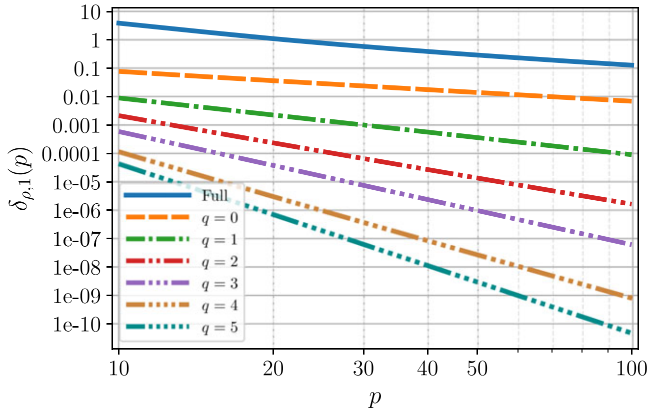

FIGURE 7 Delayed oscillator convergence using the parameters $A = 1$, $B = 0.1$. The order q refers to the order of the Lagrange polynomial used to approximate the delayed terms. See the approximated convergence rates in Table 1

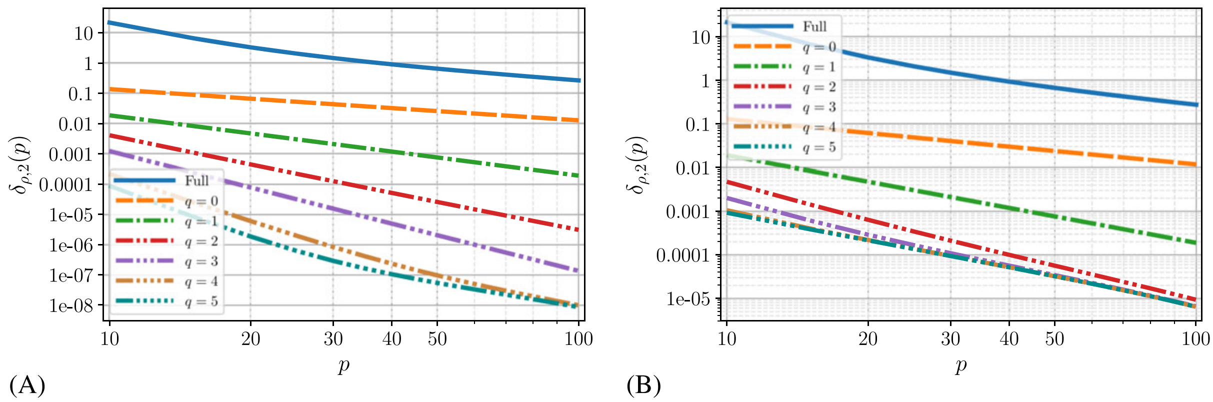

FIGURE 8 Delayed oscillator (58) convergence properties for the spectral radius of **H** with parameters $A = 1.0$, $B = 0.1$, $\beta = 0.1$ and stochastic proportional coefficient (A) $\alpha = 0$ and (B) $\alpha = 0.1$. The order q refers to the order of the Lagrange polynomial used to approximate the delayed terms. The corresponding convergence rates are given numerically in Table 1

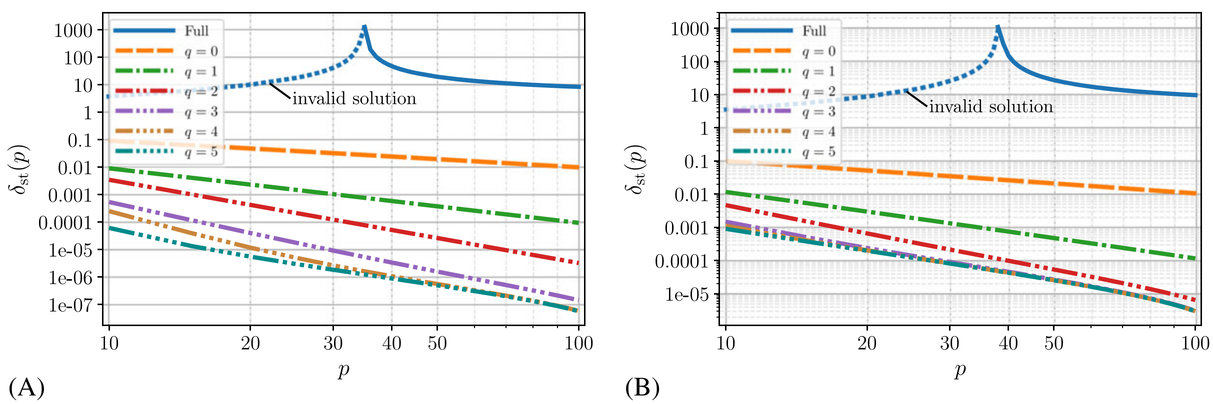

FIGURE 9 Delayed oscillator (58) convergence properties for the stationary second moment $M_{st,11}$ with parameters $A = 1.0$, $B = 0.1$, $\beta = 0.1$ and stochastic proportional coefficient (A) $\alpha = 0$ and (B) $\alpha = 0.1$. The dashed line refers to an invalid (negative) stationary solution. The order q refers to the order of the Lagrange polynomial used to approximate the delayed terms. The corresponding convergence rates are given numerically in Table 1

TABLE 1 Convergence rate r_c of the different discretization methods, by fitting $\delta(p) = Kp^{-r_c}$ function for resolution parameter $p > 50$ (FD, full-discretization method; SSD, stochastic semi-discretization method)

Test case	Quantity	Fig.	FD	*n*th-order SSD					
				0	1	2	3	4	5
Stochastic Hayes	$\delta_{\rho,2}(p)$	4A	1.03	1.02	0.97	1.00	1.00	1.00	1.00
	$\delta_{\mathrm{st}}(p)$	4B	1.08	2.00	0.97	1.01	0.99	1.00	1.00
Delayed oscillator	$\delta_{\rho,1}(p)$	7	1.18	1.02	2.00	3.06	4.00	5.08	6.00
	$\delta_{\rho,2}(p),\ \alpha = 0.0$	8A	1.28	1.01	2.00	3.06	3.95	5.23	5.71
	$\delta_{\mathrm{st}}(p),\ \alpha = 0.0$	9A	0	0.99	2.00	3.02	3.42	3.14	3.01
	$\delta_{\rho,2}(p),\ \alpha = 0.1$	8B	1.28	1.02	2.01	2.59	2.38	2.32	2.32
	$\delta_{\mathrm{st}}(p),\ \alpha = 0.1$	9B	0	0.99	2.04	2.99	2.97	2.95	2.95

It can be seen in Figure 7 that the first moment's spectral radius behaves as expected[27]: the increase of both the interpolation order of the delayed states and the resolution parameter p decreases the error $\delta_{\rho,1}$; the convergence rate r_c increases by 1 as the interpolation order increases by 1. The spectral radius of the second moment mapping matrix behaves similarly as it can be observed in Figure 8 for two different cases. In case of no multiplicative noise at the actual time t (that is for $\alpha = 0$ in Figure 8A), the one-by-one increase of the interpolation order leads to the increase of the convergence rate only by 1 (instead of 2) due to the magnitude $\sqrt{\mathrm{d}t}$ property of the Wiener process.[11] However, it can be concluded from Figure 8B that the convergence of the spectral radius of the second moment is limited for $\alpha \neq 0$ due to the error introduced by the multiplicative noise term $\boldsymbol{\alpha}\mathbf{e}^{\mathbf{A}s}\mathbf{x}_{t_n}\mathrm{d}W_s$ (see details in Appendix B.1). In this case, the use of a maximum second-order interpolation of the delayed terms is suggested since the increase in calculation time is not accompanied by significant improvement in accuracy.

In Figure 9, the convergence of the stationary second moment $M_{\mathrm{st},11} = \lim_{t\to\infty}\langle x_t^2\rangle$ can be seen. One can observe that the results gained with the full-discretization method is divergent for $p \lesssim 35 - 38$ (see the dashed line). Here, $\rho(\mathbf{H}) > 1$ implies that Equation (21) does not hold any more because it leads to a negative second moment:

$$\lim_{n\to\infty}\langle y_{n-j,i}^2\rangle < 0, \quad \forall j = 0,1,2\ldots p \text{ and } \forall i = 1,2,\ \ldots,(p+1)d, \tag{66}$$

meaning that there exists no stationary distribution of the unstable approximation. The fast convergence is especially important near the stability boundaries; this is crucial if the original system is stable but the low resolution approximate mapping is still unstable, like in the case explained above for full discretization. Consequently, the calculated (invalid) stationary second moment of the approximation first converges to $-\infty$ as the resolution p is increased, leading to a seemingly ill-working numerical method. However, the expected convergence rate can be reached in these cases for very large and costly discretization resolution p, as shown in Table 1.

6 | CONCLUSION

In the literature, the most commonly used method to determine the stationary behavior of a general stochastic linear delayed equation is conducted via time-domain simulations. To determine the stability charts in the parameter space, one has to use a large number of MC simulations, which still lead to fractal-like nonsmooth stability boundaries.[14] The semi-discretization method described in this paper provides an efficient analysis of the stationary behavior of delayed systems subjected to additive and multiplicative noise acting both at the present and at the delayed states of the system. Apart of the calculation of the stationary first and second moments, the dominant characteristic exponents of these moments are easily approximated. The convergence properties of this powerful method were demonstrated for the stochastic Hayes equation, where the analytical results are available.

The superior convergence rate of the stochastic semi-discretization over the full discretization was shown also for the stochastic delayed oscillator, which is an essential model in engineering applications, like machine tool vibrations. The efficiency of the SSD is especially important for multiple state variable systems ($d \gg 1$) since the number of the elements of the discretization matrix of the second moment mapping is proportional to $\mathcal{O}\left((pd)^4\right)$. Due to the rapid growth of the discretization matrix for a high resolution approximation, it is computationally challenging to calculate the largest eigenvalue of these very sparse matrices, for which the computational complexity is approximately $\mathcal{O}\left((pd)^{2\times 2.4}\right)$.[33] Furthermore, the solution of the linear system of Equation (21) for the stationary second moment not only needs high CPU time but also requires large amount of memory.

Due to the complexity of the general implementation of the proposed SSD method, an open source Julia[22] package is implemented under the name of *StochasticSemiDiscretizationMethod.jl* (https://github.com/HTSykora/StochasticSemiDiscretizationMethod.jl). Hopefully this easy-to-use package allows other researchers and practicing engineers to apply SSD in the broad field of stochastic delay systems.

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APPENDIX A

GENERALIZATION FOR MULTIPLE DELAYS AND INDEPENDENT NOISE SOURCES

To apply the semi-discretization method for multiple delays and independent white-noise sources, one has to use the form

$$\mathrm{d}\mathbf{x}_t = \left(\mathbf{A}\mathbf{x}_t + \sum_{o=1}^{v}\mathbf{B}^o\mathbf{x}_{t-\tau^o}\right)\mathrm{d}t + \sum_{m=1}^{w}\left(\boldsymbol{\sigma}^m + \boldsymbol{\alpha}^m\mathbf{x}_t + \sum_{o=1}^{v}\boldsymbol{\beta}^{o,m}\mathbf{x}_{t-\tau^o}\right)dW_t^m, \tag{A1}$$

where $\tau^o, o = 1, \dots, v$ are the time delays, $v \in \mathbb{N}^+$ is the number of time delays, and without the loss of generality, assume that $\tau^1 < \tau^2 < \dots < \tau^v$. Furthermore, $w \in \mathbb{N}^+$ is the number of noise sources and W_t^m, $m = 1, \dots, w$ are the Wiener processes to model the Gaussian white noises, that is, $\mathbb{E}\left(\mathrm{d}W_t^i\mathrm{d}W_t^j\right) = \mathrm{d}t\,\delta_{ij}\mathbb{1}(t = s)$ with δ_{ij} standing for the Kronecker symbol. The system in Equation (A1) can be transformed into the following mapping:

$$\mathbf{y}_{n+1} = \left(\mathbf{F} + \sum_{m=1}^{w}\mathbf{G}^m\right)\mathbf{y}_n + \sum_{m=1}^{w}\mathbf{g}^m, \tag{A2}$$

where $\mathbf{F}$ is the deterministic mapping matrix, whereas $\mathbf{G}^m$, $\mathbf{g}^m$ are the stochastic mapping matrices and additive noise vectors, respectively, each corresponding to the mth noise source. To investigate the first moment stability, one should investigate the spectral radius of $\mathbf{F}$ as described in Equation (11). The second moment stability is investigated with the mapping

$$M_{ij}(n+1) = \left(F_{il}F_{jk} + \sum_{m=1}^{w}\left\langle G_{il}^m G_{jk}^m\right\rangle\right)M_{lk}(n) + \sum_{m=1}^{w}\left(\left\langle G_{ik}^m g_j^m + G_{jk}^m g_{n,i}^m\right\rangle\right)\langle y_{n,k}\rangle + \sum_{m=1}^{w} g_i^m g_j^m \tag{A3}$$

that is the generalized form of Equation (14). In this case, however, the semi-discretization method requires the introduction of different delay resolutions

$$r^o = \mathrm{int}\left(\frac{\tau^o}{\Delta t}\right) \tag{A4}$$

for each time delay. The dimension $(p+1)d$ of the generalized mapping (A2) is determined by $p = \max_o\{r^o\} = r^v$.

APPENDIX B

IMPROVING THE CONVERGENCE PROPERTIES OF SEMI-DISCRETIZATION

B.1 | Improving the accuracy of the semi-discretization in the presence of multiplicative noise

Consider the following linear SDE:

$$\mathrm{d}\mathbf{x}_t = \mathbf{A}\mathbf{x}_t\mathrm{d}t + \boldsymbol{\alpha}\mathbf{x}_t\mathrm{d}W_t. \tag{B1}$$

In the general d dimensional case, it has no explicit solution.[19] In order to construct an analytical solution that improves the accuracy of our numerical method, the state vector $\mathbf{x}_t$ in the multiplicative term $\boldsymbol{\alpha}\mathbf{x}_t\mathrm{d}W_t$ is approximated by its expected value $\langle x_t\rangle$. Then,

$$\langle \mathrm{d}\mathbf{x}_t\rangle = \mathbf{A}\langle \mathbf{x}_t\rangle\mathrm{d}t \quad \Rightarrow \quad \langle \mathbf{x}_t\rangle = e^{\mathbf{A}t}\mathbf{x}_0\,. \tag{B2}$$

Similarly, for the approximated linear SDE,

$$\mathrm{d}\mathbf{x}_t = \mathbf{A}\mathbf{x}_t\mathrm{d}t + \boldsymbol{\alpha}e^{\mathbf{A}t}\mathbf{x}_0\mathrm{d}W_t \quad \Rightarrow \quad \mathbf{x}_t = e^{\mathbf{A}t}\left(\mathbf{I} + \int_{t_0}^{t} e^{-\mathbf{A}s}\boldsymbol{\alpha}e^{\mathbf{A}s}\mathrm{d}W_s\right)\mathbf{x}_0\,. \tag{B3}$$

To demonstrate the advantage of this approximation, the following scalar example is used:

$$\mathrm{d}x_t = Ax_t\mathrm{d}t + \alpha x_t\mathrm{d}W_t, \tag{B4}$$

which has the following exact solution[19,26] for one discretization time step Δt:

$$x_{t+\Delta t} = x_t \exp\left(\left(A - \frac{\alpha^2}{2}\right)\Delta t + \alpha\left(W_{t+\Delta t} - W_t\right)\right), \tag{B5}$$

from which the first and second moments are[19,26]

$$\langle x_{t+\Delta t}\rangle = e^{A\Delta t}\langle x_t\rangle, \tag{B6}$$

$$\langle x_{t+\Delta t}x_{t+\Delta t}\rangle = e^{2A\Delta t + \alpha^2\Delta t}\langle x_t x_t\rangle. \tag{B7}$$

When the approximation of Equation (B4) is considered without the first moment correction in the multiplicative term, we have

$$\mathrm{d}\tilde{x}_s = A\tilde{x}_s\mathrm{d}s + \alpha\tilde{x}_t\mathrm{d}W_s \quad \text{where} \quad s \in [t, t+\Delta t)\,, \tag{B8}$$

which has the following solution, the first and second moment evolutions:

$$\tilde{x}_{t+\Delta t} = \tilde{x}_t e^{A\Delta t}\left(1 + \int_{t}^{t+\Delta t} e^{-As}\alpha\mathrm{d}W_s\right), \tag{B9}$$

$$\langle \tilde{x}_{t+\Delta t}\rangle = \langle \tilde{x}_t\rangle e^{A\Delta t}, \tag{B10}$$

$$\langle \tilde{x}_{t+\Delta t}\tilde{x}_{t+\Delta t}\rangle = \langle \tilde{x}_t\tilde{x}_t\rangle\left(e^{2A\Delta t} + \frac{\alpha^2\left(e^{2A\Delta t} - 1\right)}{2A}\right), \tag{B11}$$

respectively. When the first moment correction is applied in the approximation, the following SDE is obtained:

$$\mathrm{d}\bar{x}_s = A\bar{x}_s\mathrm{d}s + \alpha\bar{x}_t e^{As}\mathrm{d}W_s \quad \text{where} \quad s \in [t, t+\Delta t)\,, \tag{B12}$$

and the corresponding solutions are

$$\bar{x}_{t+\Delta t} = \bar{x}_t e^{A\Delta t}\left(1 + \int_{t}^{t+\Delta t} e^{-As}\alpha e^{As}\mathrm{d}s\right), \tag{B13}$$

$$\langle \bar{x}_{t+\Delta t}\rangle = \langle \bar{x}_t\rangle e^{A\Delta t}, \tag{B14}$$

$$\langle \bar{x}_{t+\Delta t}\bar{x}_{t+\Delta t}\rangle = \langle \bar{x}_t\bar{x}_t\rangle e^{2A\Delta t}\left(1 + \alpha^2\Delta t\right). \tag{B15}$$

To compare the quality of the different approximations of the second moment, the Taylor series of the second moments is used for the exact solution (B7)

$$\langle x_{t+\Delta t} x_{t+\Delta t} \rangle = \langle x_t x_t \rangle \left(1 + \left(2A + \alpha^2\right) \Delta t + \left(2A^2 + 2A\alpha^2 + \frac{\alpha^4}{2} \right) \Delta t^2 \right) + \mathcal{O}(\Delta t^3), \tag{B16}$$

for the noncorrected approximating solution (B11)

$$\langle \tilde{x}_{t+\Delta t} \tilde{x}_{t+\Delta t} \rangle = \langle \tilde{x}_t \tilde{x}_t \rangle \left(1 + \left(2A + \alpha^2\right) \Delta t + \left(2A^2 + A\alpha^2\right) \Delta t^2\right) + \mathcal{O}(\Delta t^3), \tag{B17}$$

and for the corrected approximating solution (B15)

$$\langle \bar{x}_{t+\Delta t} \bar{x}_{t+\Delta t} \rangle = \langle \bar{x}_t \bar{x}_t \rangle \left(1 + \left(2A + \alpha^2\right) \Delta t + \left(2A^2 + 2A\alpha^2\right) \Delta t^2\right) + \mathcal{O}(\Delta t^3). \tag{B18}$$

From the error

$$\langle x_{t+\Delta t} x_{t+\Delta t} \rangle - \langle \tilde{x}_{t+\Delta t} \tilde{x}_{t+\Delta t} \rangle = \left(A\alpha^2 + \frac{1}{2}\alpha^4 \right) \langle x_t x_t \rangle \Delta t^2 + \mathcal{O}(\Delta t^3) \tag{B19}$$

of the noncorrected approximation and from the error

$$\langle x_{t+\Delta t} x_{t+\Delta t} \rangle - \langle \bar{x}_{t+\Delta t} \bar{x}_{t+\Delta t} \rangle = \frac{1}{2}\alpha^4 \langle x_t x_t \rangle \Delta t^2 + \mathcal{O}(\Delta t^3) \tag{B20}$$

of the first moment corrected approximation, it can be seen that the two methods have the same order in Δt. However, the multiplicative term α is typically only a small fraction of the term A, so the error of the corrected approximation in α is two orders of magnitude smaller. Note that this extension has negligible effect on the computational time of the numerically determined integrals in Equation (31).

B.2 | Higher order semi-discretization

To improve the accuracy of the approximation of the mapping, Lagrange polynomials are used to approximate the delayed term.[27] The term higher order refers to the order of the Lagrange polynomial used to approximate the delayed terms. This leads to additional terms in both the deterministic and the stochastic mapping matrices. Introducing the delay resolution r by

$$r = \operatorname{int}\left(\frac{\tau}{\Delta t} + \frac{q}{2} \right), \tag{B21}$$

where q is the order of the Lagrange polynomial

$$\mathbf{L}^{(q)}(t-\tau) = \sum_{k=0}^{q} \left(\prod_{l=0, l\neq k}^{q} \frac{t - \tau - (n + l - r)\Delta t}{(k-l)\,\Delta t} \right) \mathbf{x}_{t-(n+k-r)\Delta t}. \tag{B22}$$

Using this, the approximation of Equation (8) leads to

$$\mathrm{d}\mathbf{x}_s = \left(\mathbf{A}\mathbf{x}_s + \mathbf{B}\,\mathbf{L}^{(q)}(s-\tau)\right) \mathrm{d}s + \left(\boldsymbol{\alpha} e^{\mathbf{A}s}\mathbf{x}_{t_n} + \boldsymbol{\beta}\,\mathbf{L}^{(q)}(s-\tau) + \boldsymbol{\sigma}\right) \mathrm{d}W_s, \quad \text{where} \ \ s \in [t_n, t_{n+1}), \tag{B23}$$

with the discretized form

$$x_{t+\Delta t} = (\mathbf{P} + \mathbf{Q})\,\mathbf{x}_t + \sum_{k}^{q} (\mathbf{R_k} + \mathbf{S_k})\,\mathbf{x}_{t-\tau+k\Delta t} + \mathbf{w}. \tag{B24}$$

Example: first-order semi-discretization

In case of choosing $q = 1$, one approximates the delayed terms with a first-order polynomial. In terms of discretization matrices, this leads to the following deterministic matrices:

$$\mathbf{P} = e^{\mathbf{A}\Delta t}, \tag{B25}$$

$$\mathbf{R}_0 = \left(\mathbf{A}^{-1} + \frac{1}{\Delta t} \left(\mathbf{A}^{-2} - (\tau - (r-1)\Delta t)\,\mathbf{A}^{-1} \right) \left(\mathbf{I} - e^{\mathbf{A}\Delta t} \right) \right) \mathbf{B}, \tag{B26}$$

$$\mathbf{R}_1 = \left(-\mathbf{A}^{-1} + \frac{1}{\Delta t} \left(-\mathbf{A}^{-2} + (\tau - r\Delta t)\,\mathbf{A}^{-1} \right) \left(\mathbf{I} - e^{\mathbf{A}\Delta t} \right) \right) \mathbf{B}, \tag{B27}$$

stochastic matrices

$$\mathbf{Q} = \int_{t_n}^{t_{n+1}} e^{\mathbf{A}(t-s)} \boldsymbol{\alpha} e^{\mathbf{A}s} \mathrm{d}W_s, \tag{B28}$$

$$\mathbf{S}_0 = \int_{t_n}^{t_{n+1}} e^{\mathbf{A}(t-s)} L_0^{(1)}(s) \boldsymbol{\beta} \mathrm{d}W_s, \tag{B29}$$

$$\mathbf{S}_1 = \int_{t_n}^{t_{n+1}} e^{\mathbf{A}(t-s)} L_1^{(1)}(s) \boldsymbol{\beta} \mathrm{d}W_s, \tag{B30}$$

and stochastic additive vector

$$\mathbf{w} = \int_{t_n}^{t_{n+1}} e^{\mathbf{A}(t-s)} \boldsymbol{\sigma} \mathrm{d}W_s. \tag{B31}$$

Choosing $p = r$, the discretization matrices and stochastic additive vector in Equation (9) assume the following structure:

$$\mathbf{F} = \begin{pmatrix} \mathbf{P} & \mathbf{0} & \dots & \mathbf{R}_1 & \mathbf{R}_0 \\ \mathbf{I} & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \dots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{I} & \mathbf{0} \end{pmatrix}, \quad \mathbf{G} = \begin{pmatrix} \mathbf{Q} & \mathbf{0} & \dots & \mathbf{S}_1 & \mathbf{S}_0 \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \end{pmatrix}, \quad \mathbf{g} = \begin{pmatrix} \mathbf{w} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{pmatrix}. \tag{B32}$$

APPENDIX C

STRUCTURE OF THE SECOND MOMENT GENERATOR MATRIX

To show the structure of matrix $\mathbf{H}$ using the arrangement of the elements of the second moment matrix $\mathbf{M}_n$ provided in Equations (19), matrix $\mathbf{H}$ is partitioned into terms originating from matrices $\mathbf{F}$ and $\mathbf{G}$:

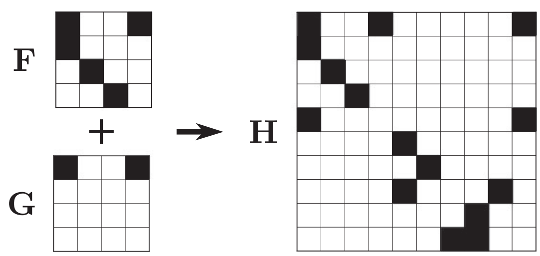
$$\mathbf{H} = \mathbf{H}_\mathbf{F} + \mathbf{H}_\mathbf{G}. \tag{C1}$$

Note that both $\mathbf{H}_\mathbf{F}$ and $\mathbf{H}_\mathbf{G}$ have identical structures, but since $\mathbf{G}$ is stochastic, the expectation value operator has to be used: if $H_{\mathbf{F},11} = F_{11}^2$, then $H_{\mathbf{G},11} = \langle G_{11}^2 \rangle$.

In the case of $\mathbf{F}, \mathbf{G} \in \mathbb{R}^{3\times3}$ $\mathbf{H}_\mathbf{F}$ is the following:

$$\mathbf{H}_\mathbf{F} = \left(\begin{array}{ccc|cc|c} F_{11}^2 & F_{12}^2 & F_{13}^2 & 2F_{11}F_{12} & 2F_{12}F_{13} & 2F_{11}F_{13} \\ F_{21}^2 & F_{22}^2 & F_{23}^2 & 2F_{21}F_{22} & 2F_{22}F_{23} & 2F_{21}F_{23} \\ F_{31}^2 & F_{32}^2 & F_{33}^2 & 2F_{31}F_{32} & 2F_{32}F_{33} & 2F_{31}F_{33} \\ \hline F_{11}F_{21} & F_{12}F_{22} & F_{13}F_{23} & F_{11}F_{22}+F_{12}F_{21} & F_{12}F_{23}+F_{13}F_{22} & F_{11}F_{23}+F_{13}F_{21} \\ F_{21}F_{31} & F_{22}F_{32} & F_{23}F_{33} & F_{21}F_{32}+F_{22}F_{31} & F_{22}F_{33}+F_{23}F_{32} & F_{21}F_{33}+F_{23}F_{31} \\ \hline F_{11}F_{31} & F_{12}F_{32} & F_{13}F_{33} & F_{11}F_{32}+F_{12}F_{31} & F_{12}F_{33}+F_{13}F_{32} & F_{11}F_{33}+F_{13}F_{31} \end{array} \right). \tag{C2}$$

The schematic structures of the deterministic, stochastic, and second moment mapping matrices $\mathbf{F}$, $\mathbf{G}$, and $\mathbf{H}$, respectively, are presented for zeroth-order semi-discretization with $p = 3$ (see Equations (18)-(20)).



APPENDIX D

ZEROTH-ORDER SEMI-DISCRETIZATION OF THE STOCHASTIC HAYES EQUATION

To illustrate the construction of the second moment mapping defined in Equation (18), the M_{11} element of the second moment mapping of the stochastic Hayes equation is given using two separate approaches. First, the original stochastic Hayes equation from Equation (34)) is considered:

$$\mathrm{d}x_t = Ax_t\mathrm{d}t + \beta x_{t-1}\mathrm{d}W_t + \sigma\mathrm{d}W_t. \tag{D1}$$

Its zeroth-order semi-discretized form similar to Equation (22) is

$$\mathrm{d}x_t = Ax_t\mathrm{d}t + \beta x_{t_{n-p}}\mathrm{d}W_t + \sigma\mathrm{d}W_t, \quad \text{where} \quad t \in [t_n, t_{n+1}\Delta t]. \tag{D2}$$

The calculated stochastic mapping matrices and vector using Equations (24), (25), (26), and (27)

$$\mathbf{F} = \begin{pmatrix} F_{11} & 0 & \dots & 0 \\ 1 & 0 & \dots & 0 \\ 0 & \ddots & & \vdots \\ 0 & \dots & 1 & 0 \end{pmatrix}, \qquad \mathbf{G} = \begin{pmatrix} 0 & 0 & \dots & G_{1,p+1} \\ 0 & 0 & \dots & 0 \\ 0 & \ddots & & \vdots \\ 0 & \dots & & 0 \end{pmatrix}, \qquad \mathbf{g} = \begin{pmatrix} g_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \tag{D3}$$

where

$$F_{11} = e^{A\Delta t}, \qquad G_{1,p+1} = \beta \int_{t_n}^{t_{n+1}} e^{A(t_{n+1}-s)}\mathrm{d}W_s, \qquad g_1 = \sigma \int_{t_n}^{t_{n+1}} e^{A(t_{n+1}-s)}\mathrm{d}W_s. \tag{D4}$$

The elements of the second moment matrix which will be used are the following:

$$\left\langle x_{t_n}x_{t_n} \right\rangle := M_{11}(n), \qquad \left\langle x_{t_{n-p}}x_{t_{n-p}} \right\rangle := M_{p+1,p+1}(n). \tag{D5}$$

Approximating the moment evolution using the semi-discretization matrices and the stepping definitions from Equation (18) leads to the mapping for $M_{11}(n + 1)$:

$$M_{11}(n+1) = F_{11}^2 M_{11}(n) + G_{1,p+1}^2 M_{p+1,p+1}(n) + 2G_{1,p+1}g_1\langle x_{t_{n-p}}\rangle + g_1^2. \tag{D6}$$

Substituting the matrix elements from Equation (D3) into Equation (D6) and performing the Itô integral as in Equations (31)-(33) lead to the mapping

$$M_{11}(n+1) = e^{2A\Delta t}M_{11}(n) + \frac{\beta^2\left(e^{2A\Delta t}-1\right)}{2A}M_{p+1,p+1}(n) + \frac{\beta\sigma\left(e^{2A\Delta t}-1\right)}{A}\langle x_{(n\Delta t-p)\Delta t}\rangle + \frac{\sigma^2\left(e^{2A\Delta t}-1\right)}{2A}. \tag{D7}$$

To show the correctness of the above result, the second approach is used; the analytically derived second moment Equation (46) is considered

$$\dot{M}(t) = 2AM(t) + \beta^2 M(t-1) + 2\beta\sigma\langle x_{t-1}\rangle + \sigma^2. \tag{D8}$$

For this particular equation, the application of the zeroth-order semi-discretization leads to

$$\dot{M}(t) \approx 2AM(t) + \beta^2 M\left(t_{n-p}\right) + 2\beta\sigma\langle x_{t_{n-p}}\rangle + \sigma^2 \quad \text{where} \quad t \in [t_n, t_{n+1}]. \tag{D9}$$

Solving the above linear differential equation on the time interval $t \in [t_n, t_{n+1}]$ leads to:

$$M\left(t_{n+1}\right) = e^{2A\Delta t}M\left(t_n\right) + \frac{\beta^2\left(e^{2A\Delta t}-1\right)}{2A}M\left(t_{n-p}\right) + \frac{\beta\sigma\left(e^{2A\Delta t}-1\right)}{A}\langle x_{(n\Delta t-p)\Delta t}\rangle + \frac{\sigma^2\left(e^{2A\Delta t}-1\right)}{2A}, \tag{D10}$$

which is the same compared with Equation (D7) since

$$M(t_n) := M_{11}(n), \qquad M(t_{n-p}) := M_{p+1,p+1}(n). \tag{D11}$$